Undergraduate Thesis

Final Report (Thesis) Specification

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- **Weight of the Final Report – (which is Thesis itself):** Please refer to the course description where the weight of the Thesis is indicated.
- Submission of the Thesis:

  (i) FIRST, please submit the Thesis to your supervisor.
  (ii) THEN, please submit it to OWL as per the specs given in the OWL assignment.

Format

1. Please submit your thesis in either MS WORD or PDF.
2. Page header:
   - Page number (formatted as Page X of Y) on the top left corner of every page.
   - You Last name, First name on the top right corner of every page.
   - Pages must be SINGLE spaced.
   - All text must be in 12 pt. Times New Roman font.
   - Title page – all centered:
     - Project title
     - Subtitle **"Final Report – Undergraduate Thesis"**

   - Your First and Last name -- (official UWO name, please!)
   - Course name and number
   - Department of Computer Science
   - Western University
   - Date
   - Project supervisor name and his/her Department
   - Course coordinator: Prof. Nazim Madhavji, Dept. of Computer Science.

 General remarks:

- Give examples to support your conceptual or theoretical points to improve understandability.
- Please adhere to the length limits given (tolerance of 10-15% of the limit)
- Please use the Appendix to describe material that does not fit in the space limits given here.

 Citation and References:

Please use the APA Style. (e.g., (Crockatt, 1995)). See:
http://pitt.libguides.com/c.php?g=12108&p=64730

 Length requirements:

- The length of each section is indicated.

 Glossary

- Create a section on Glossary BEFORE the structured abstract.
- List acronyms and their expanded definitions.
- In the report body, at the *first* point of use of an acronym, please include the expanded version in the text and its acronym in brackets "(….)"

 Structured Abstract**:** [up to 300 words]

- Write a structured abstract as in PR. Details are NOT repeated here.
- Don't simply copy PR abstract here; in particular, the final thesis is expected to have solid results and conclusion and updates to research methods used, etc.

 Table of Contents

After the structured abstract and before the report sections, please include here the Table of Contents

1) Introduction [up to 1.5 pages.]

 Include the following:

- general background and problem description,
- focus of this thesis
- key research questions addressed,
- key results,
- novelty and significance,
- report structure.

TIP:

- Write Introduction last after the rest of the report has been written because it becomes much easier to introduce the whole report.
- In "general background and problem description" (see above), it is important to cite related references.

2) Background and Related Work [Up to 2 pages]

- A sequence of **subsections numbered** 2.1, 2.2, 2.3, etc. each with a **subsection title**:
    - Each subsection contains tightly related points for that subsection (includes citations).
- **Analysis and Research Gap** [0.5 pages max.]:
  Analysis: Compare and contrast research within and across the subsections 2.1, 2.2, 2.3, etc.
  Gap: Describe the Research Gap and how this Gap relates to the core work described in the thesis.

TIP: See the guideline table below for writing the analysis sub-section (Mary Shaw, ICSE 2003):

Proceedings of the 25th International Conference on Software Engineering (ICSE'03)

Explain the relation to other work clearly …

Awful	▼	The galumphing problem has attracted much attention [3,8,10,18,26,32,37]
Bad	▼	Smith [36] and Jones [27] worked on galumphing.
Poor	▼	Smith [36] addressed galumphing by blitzing, whereas Jones [27] took a flitzing approach.
Good	▲	Smith's blitzing approach to galumphing [36] achieved 60% coverage [39]. Jones [27] achieved 80% by flitzing, but only for pointer-free cases [16].
Better	▲	Smith's blitzing approach to galumphing [36] achieved 60% coverage [39]. Jones [27] achieved 80% by flitzing, but only for pointer-free cases [16]. We modified the blitzing approach to use the kernel representation of flitzing and achieved 90% coverage while relaxing the restriction so that only cyclic data structures are prohibited.

3) Research Goal and Objectives [0.5 page max.]

- Describe the overall Goal of the thesis.
- State the objectives of your project, formatted as bullet points:

  - O1: <objective 1 text>
  - O2: <objective 2 text>

  Etc.

  Each objective should have an id. (e.g., O1, O2, etc.) such that you can make back references to them from the rest of the report (esp. from methodology, results, and conclusions). Describe how the set of objectives aim at attaining the goal.

4) Methodology [1- 1.5 pages max.]

(Note: Please consult with your supervisor for the appropriate methodology or procedures to be used for your thesis as not all the research methods are represented here). Example:

- **If "developing" a system**:
  - Describe here research and development approaches used:
    - Cite specific concepts, algorithms, techniques, tools, etc., used in creating your *solution*. **Give citations**.
    - Cite specific technologies used in your development *process*. **Give citations**.
- **If conducting an "empirical study"** (e.g., case study, scientific experiment, systematic literature review, etc.):
  - Describe the research design and empirical procedures followed.
    - **Give citations** to research design and the research procedures followed.
- **If conducting a "data science-related" study:**
  - Describe key methodological issues, e.g.: data gathering, data quality, data sampling, machine learning model training, data analysis procedures, ethical issues, etc.

5) Results [5 pages max.]
(Note: Similar to methodological issues, please consult with your supervisor for adequately describing the results of your research). Example:

- **(a) If "creating" a novel system**, give in appropriate sub-sections:
  - A schematic "**contextual diagram**" placing the overall system in the environment in which it is meant for use.
    - If your development project is a sub-part of the overall system, depict your contribution and clearly describe how it interfaces with the rest of the system.
  - Describe the **technical work** carried out, including:
    - **Key requirements** that were met.
    - **System design and architecture**, highlighting new components and design you have added vs. those that pre-existed.
      - Give details of: (i) any design or architectural/design patterns used; (ii) definitions of component interfaces; quality attributes (such as performance, usability, reliability, portability, inter-operability, security, privacy, etc.) considered in the design, giving specific design fragments to support your claim;
        - Give rationale for your design decisions.
    - **System implementation and testing**:
      - Describe any new algorithms or techniques developed.
      - Describe frameworks, platforms, libraries, algorithms,
      - languages, and tools, etc. used in developing your solution, justifying your choices.
      - Explain how the system was tested, explaining: unit testing, integration testing, regression testing, acceptance testing, etc.

- **System Validation**:
  - Explain if, and how, the implemented system was validated.
    - Describe "case studies" or scenarios conducted, including their results.
    - Show the system "operationally", step by step, through screenshots.
    - Describe any stakeholder (end-user, supervisory, etc.) feedback provided.

- **OR (b) If conducting an "empirical study"** then in appropriate sub-sections:
  - Describe the "design" of the "study"; data gathering; execution of the study; procedural or data challenges and how these were handled. See, for example:
    - Runeson, P., Höst, M.: Guidelines for conducting and reporting case study research in software engineering. *Empir Software Eng* **14**, 131–164 (2009). https://doi.org/10.1007/s10664-008-9102-8.
    - Claes Wohlin, et al.: Experimentation in Software Engineering, Springer Publishing Company, Incorporated, ISBN:978-3-642-29043-5, 2012.
  - Describe the findings of the study.

- **OR (c) If conducting a "data science related study"** then the results such as classification of data, findings from data analysis, effectiveness of various configurations of algorithms, etc.

- **Novelty of Results**:

Describe what is novel in your result.

- Compare your results with core Related Work. Describe comparative procedures followed, show comparative data, and conclude on relative novelty.

6) **Discussion** [1-2 pages max.].

- Threats to the validity of the results
  - Describe any factors that could have affected the quality of the results. Describe how such threats were mitigated and if there are any residual threats.
- Implications of the research results
  - Describe how the research results would change anything in the world. Example:
    - What is the anticipated impact on related research?
      - How could the results improve existing concepts, methods, techniques and tools or spur on new research?

- What is the anticipated impact on practice?
  - In which way would the results benefit/change practice?
- Limitations of the results
  - Describe the conditions under which your results may be limited or may not satisfy the claim you have made.
  - Can these conditions be minimised, avoided or eliminated?
- Generalisability of the results
  - Describe the extent to which the results of your research are generalisable to other situations.

7) Conclusions [0.5 page max.]

- Briefly describe the problem and the objectives of your research project.
  - Give citations.
- Highlight the key results achieved, including novelty.
- *Important*: Give back-references into the report (section number, table, figure, etc.) where the results are described in details.
- What conclusions do you draw from your work?

8) Future Work and Lessons Learnt [up to 1 page max.].

- List future work that can be conducted to enhance the described work?
- Briefly describe "significant" (new) lessons learnt from this research project that could be useful to others.
  - NOTE: lessons learnt should be novel in the literature. Do not include lessons that "you" have learnt but are NOT new in the scientific literature.

9) Acknowledgments

- Help received from others in this research (e.g., from the supervisor, other collaborators, etc.).

10) Appendix

Please use Appendix to describe material that does not fit in the space limits given for the main part of the thesis.

11) References

- Please use the APA style. See: http://pitt.libguides.com/c.php?g=12108&p=64730.